

Lecture 3 – Linear Regression

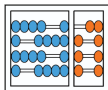
[ISLP 3.1–3.2]

MC886 – Machine Learning

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Today's summary

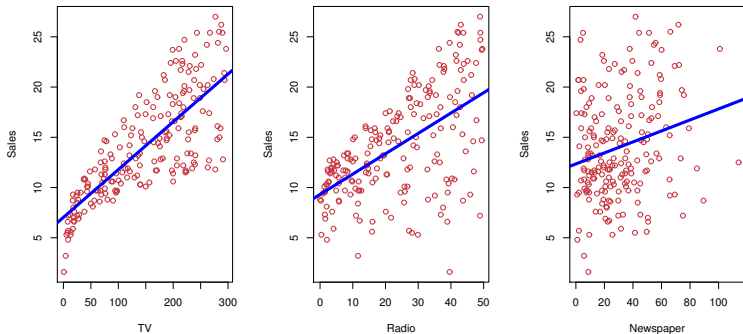
Simple linear regression

Assessing accuracy of the model

Multiple linear regression

Qualitative predictors and extensions of the linear model

The Advertising dataset



The plot displays sales, in thousands of units, as a function of TV, radio, and newspaper budgets, in thousands of dollars, for 200 different markets. Each blue line represents a simple model that can be used to predict sales using TV, radio, and newspaper, respectively

Questions regarding the Advertising data

- ❖ How to model the relationship between ad budget and sales?
- ❖ How accurately can we predict future sales?
- ❖ Is the relationship linear?
- ❖ Is there a synergy among the advertising media?

Simple linear regression

- ❖ **Assumption:**

an approximately linear relationship between X and Y

$$Y \approx \beta_0 + \beta_1 X \quad (3.1)$$

- ❖ We can regress **sales** on **TV** by fitting the model:

$$\text{sales} \approx \beta_0 + \beta_1 \text{TV}$$

- ❖ The fitted model can then be used to predict a sale $\hat{y}_i$ based on a TV ad value x_i :

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i \quad (3.2)$$

Estimating the parameters β_0 and β_1

- ❖ We'll use a dataset $\{(x_1, y_1), (x_2, y_2), \dots (x_n, y_n)\}$
- ❖ We want $\hat{\beta}_0$ and $\hat{\beta}_1$ such that the resulting line is as close as possible to the n points
- ❖ **Question:** How to define *closeness* here?
 - A possibility: to minimize the *least squares* criterion

Residual sum of squares (RSS)

- ❖ Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$
be the prediction of Y based on the i th value of X
- ❖ The i th residual is $e_i = y_i - \hat{y}_i$
(i.e., the difference between the real (y_i) and predicted ($\hat{y}_i$) values)
- ❖ The residual sum of squares (RSS) is defined as:

$$\begin{aligned}RSS &= e_1^2 + e_2^2 + \dots + e_n^2 \\&= (y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + \dots + (y_n - \hat{y}_n)^2 \\&= (y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_1)^2 + (y_2 - \hat{\beta}_0 - \hat{\beta}_1 x_2)^2 + \dots + (y_n - \hat{\beta}_0 - \hat{\beta}_1 x_n)^2\end{aligned}\tag{3.3}$$

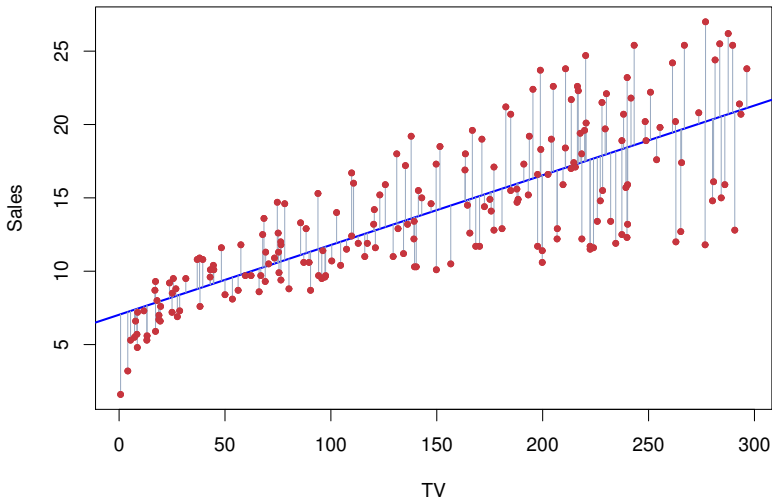
Minimizing least squares

- ❖ For simple linear regression, it holds that the least squares criterion is minimized through:

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x},\end{aligned}\tag{3.4}$$

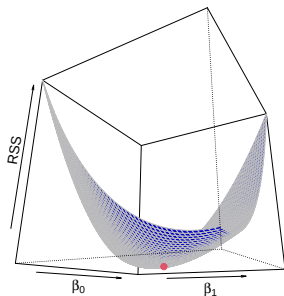
where $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ and $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

The least squares fitting for Advertising



RSS as a function of $\hat{\beta}_0$ and $\hat{\beta}_1$ in Advertising

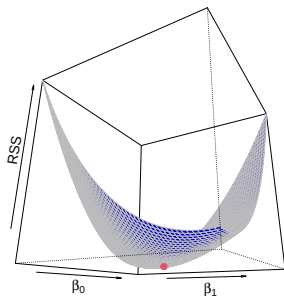
3D plot



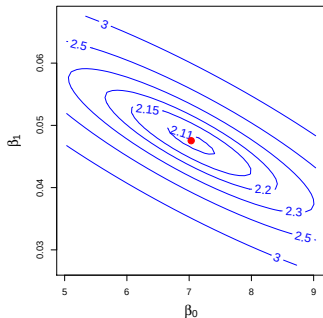
Red dots are the least squares estimates $(\hat{\beta}_0, \hat{\beta}_1)$ obtained using Eq. 3.4

RSS as a function of $\hat{\beta}_0$ and $\hat{\beta}_1$ in Advertising

3D plot



contour plot



Red dots are the least squares estimates $(\hat{\beta}_0, \hat{\beta}_1)$ obtained using Eq. 3.4

Assessing accuracy of the model

- ❖ “*To which extent does the model fit the data?*”
- ❖ We can assess that with two related quantities:
 - The *residual standard error* (RSE)
 - The R^2 statistic

Residual Standard Error (RSE)

- ❖ This quantity is given by:

$$RSE = \sqrt{\frac{1}{n-2}RSS} = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3.15)$$

- ❖ RSE is the average amount that the response will deviate from the true regression line

The R^2 statistic

- ❖ Is the proportion of variance (i.e., variability) explained in the response after the regression is performed:

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}, \quad (3.17)$$

where TSS is *the total sum of squares*: $\sum_{i=1}^n (y_i - \bar{y})^2$

- ❖ R^2 always lies between 0 and 1
- ❖ **Questions:** what does a R^2 close to 0 mean? And close to 1?

Model accuracy assessment on Advertising

Quantity	Value
RSE	3.26
R^2	0.612

Question: How to know that this performance will be satisfactory on observations that are not present in the dataset?

Train/test split

- ❖ We split dataset observations into *training* and *test* sets
- ❖ The split ratio depends on the data domain and dataset size (a typical value is 70% for training and 30% for test)
- ❖ Model training must be done using only training set
- ❖ **IMPORTANT:** by any means, do not use test data in the model learning (“leakage of test into training”)

Regression on two or more variables

- ❖ Going back to the **Advertising** problem, we also have data for ad invested on radio and newspapers
 - How to use them?
- ❖ **Idea:** to run three separate simple linear regressions
 - Let's see that idea working...

Multiple linear regression

- ❖ We extend the simple linear regression model by giving each of the p variables a separate parameter in a single model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \epsilon, \quad (3.19)$$

where X_j is the j th variable and β_j quantifies the association between it and Y

- ❖ β_j quantifies the average effect on Y of a one unit change in X_j , considering all other variables fixed
- ❖ In the **Advertising** example, we have:

$$\text{sales} = \beta_0 + \beta_1 \cdot \text{TV} + \beta_2 \cdot \text{radio} + \beta_3 \cdot \text{newspaper} + \epsilon \quad (3.20)$$

Estimating the regression parameters

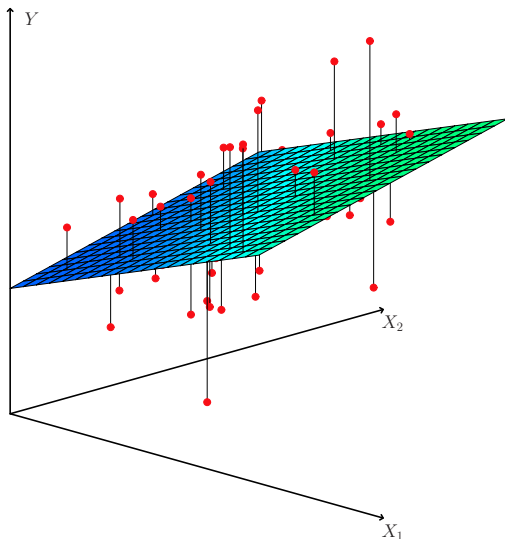
- ❖ As in simple linear regression, the parameters $\beta_0, \beta_1, \dots, \beta_p$ are unknown and must be estimated, in order to be used for predictions through the formula:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_p x_p \quad (3.21)$$

- ❖ Here we also use the same least squares approach:

$$\begin{aligned} RSS &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2} - \dots - \hat{\beta}_p x_{ip})^2 \end{aligned} \quad (3.22)$$

Least squares regression line for two variables



Some important questions in multiple linear regression

1. Is there a subset of the variables useful for predicting Y ?
2. How well does the model fit the data?

1. Deciding on important variables

- ❖ Ok, let's then try to find which subset of variables X is the most associated with Y
- ❖ Such procedure is known as *variable selection*
- ❖ There's a number of approaches to this end (we'll see some of them in a few lectures)

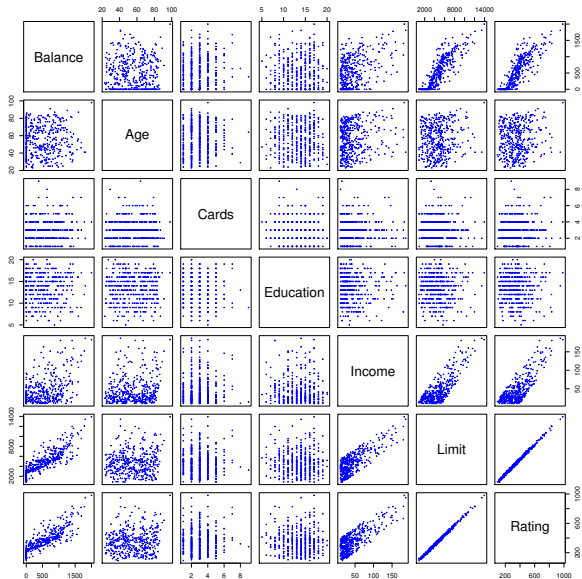
2. Model fit

- ❖ In multiple linear regression, we can evaluate the model fit with R^2 and also with the general form of the residual standard error (RSE):

$$RSE = \sqrt{\frac{1}{n - p - 1} RSS} \quad (3.25)$$

- ❖ RSE of simple linear regression is a special case where $p = 1$
- ❖ We need to run R^2 /RSE in both training and test sets. **Why?**

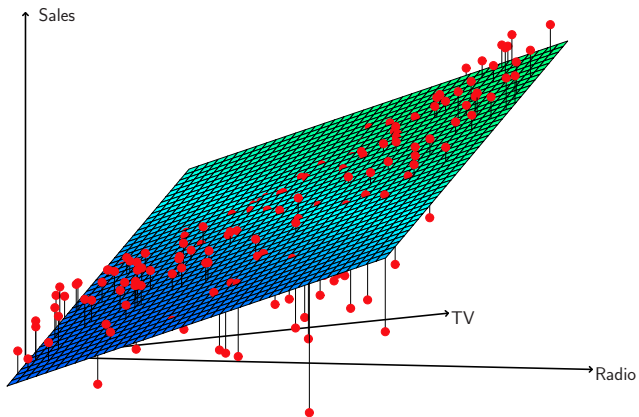
Quantitative variables in Credit dataset



And what if we have *qualitative* variables?

- ❖ **Credit** dataset has qualitative variables such as house ownership (Yes/No) and region (North, East, ...)
- ❖ **Workaround:** to create one *dummy* binary variable for each category
- ❖ This approach is also known as *one-hot encoding*

Non-linear relationships in Advertising data



Extensions of the linear model

- ❖ We can capture those non-linear interactions among variables
- ❖ For the previous example, we could propose:

$$\text{sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 (\text{radio} \times \text{TV}) + \epsilon \quad (3.33)$$

- ❖ We could further extend to a *polynomial regression*. e.g.:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \beta_4 X_1^2 + \beta_5 X_2^2 + \epsilon$$

Capturing non-linear relationships in **Auto** dataset

